NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
End Semester Examination December-2024

Paper: Economics for Engineers **Max Marks: 50**
Paper Code: HSIC-102 **Time: 3 Hours**

General Instructions

- All questions are compulsory. There is internal choice.
- Attempt the questions in the order given on the pages without skipping any

Section 1: Attempt any five (3 marks each)

1. Define sunk cost and fixed cost. How are they different?
2. Explain the concept of economies and diseconomies of scale with two factors causing them.
3. What are the two major types of taxes? Provide examples.
4. Define marginal cost and average cost. Using the table below, calculate the missing values for average cost and marginal cost:

Output (Units)	Total Cost (₹)	Average Cost (₹)	Marginal Cost (₹)
1	50	?	50
2	90	?	?
3	120	?	?
4	160	?	?

5. The price of X is ₹50 per unit and Y is ₹25 per unit. If a consumer has a budget of ₹500, how many units of X and Y can the consumer buy? Illustrate the budget line and explain the changes if the price of X increases to ₹100 or if the budget decreases to 200.
6. Explain the concept of price elasticity of supply. If the price of a product rises from ₹50 to ₹150 and quantity supplied increases by 40%, calculate the price elasticity of supply.
7. What is an exchange rate, and why is it important?

Section 2: Attempt any five (5 marks each)

8. Define average product and marginal product. Graphically and mathematically explain their relationship.
9. Explain the law of demand and discuss factors affecting the demand curve.
10. A company is considering an investment project requiring an initial outlay of $500,000. The project is expected to generate the following cash flows:
    - Year 1: $120,000
    - Year 2: $140,000
    - Year 3: $160,000
    - Year 4: $180,000
    - Year 5: $200,000

Calculate the Net Present Value (NPV) at a discount rate of 10% and advise whether the project should proceed.

11. What is the law of equi-marginal utility? Explain it using an indifference curve and a budget line.
12. A consumer has a monthly budget of ₹800. The price of bread is ₹40 per unit, and the price of milk is ₹20 per liter. If the utility function is $U(B, M) = 2B + M^2$, determine the optimal consumption bundle to maximize utility.
13. Shed light on two major economic problems (poverty and unemployment) in India and discuss measures to control them.

Section 3: Attempt any one (10 marks each)

14. Discuss the key policy reforms initiated in 1991. Why were they introduced, and how have they impacted the Indian economy in the long run?
15. Inflation remains high while economic growth slows in India. Comment on this scenario and suggest how the government and RBI can address these challenges using policy tools.